

Old School Code, New School Learning

The Technical Powerhouse

Don't let the playful pixel art fool you—Farm Counting Fiesta is a fully functional educational video game written entirely in 8086 Assembly Language. Running as a bare-metal, raw .COM executable within VGA Mode 13h, the game directly manipulates CPU registers and the linear frame buffer at memory segment 0xA000

To achieve smooth, flicker-free rendering on retro hardware, the engine utilizes custom Double Buffering and waits for vertical retrace synchronization before flipping frames. It's complex, classic low-level computer science engineering packaged into an interactive experience.

Early Childhood Benefits



Cognitive Math Skills



Motor Coordination



Zero Loading Screens

FARM COUNTING FIESTA

GROUP 1 | BSCS 2-4

COMPUTER ARCHITECTURE AND
ORGANIZATION (CSC 0222-4)

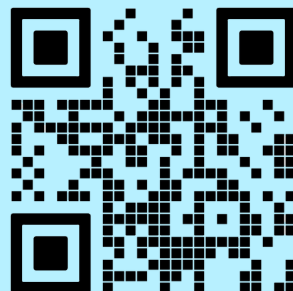
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PLAY NOW!



FARM COUNTING FIESTA

100% ASSEMBLY POWERED

GROUP 1

MEET THE FARM FRIENDS

The Gameplay Loop

→ COUNT

The engine randomly renders hand-crafted pixel animal sprites (chickens, cows, or pigs) on the screen.

→ INPUT

The player types their numeric answer using a child-friendly interface.

→ VERIFY

Pressing ENTER parses the input instantly, triggering reactive audio and visual scoring states.

Programmatic Difficulties

EASY (1-5 Animals)

MEDIUM (1-8 Animals)

HARD (1-10 Animals)

HIGH-FIDELITY IN LOW-LEVEL CODE

Secure Authentication System



- **Local Accounts:** Supports unique username and password creation directly from the interface.
- **Custom Database:** Account data is read and written dynamically using low-level DOS file I/O operations to a secure local ACCOUNTS.DAT system database.

Persistent Token & Score Tracking



The Token Economy

Players earn Barn Tokens for correct answers (tracked carefully via half-token integer values in the backend code)



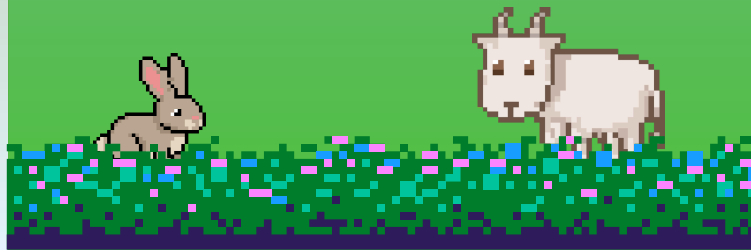
Live Leaderboards

High scores are automatically sorted using an optimized assembly bubble sort algorithm.



Data Persistence

Top scores are saved permanently across sessions in SINGLE_SCORES.DAT and DUEL_SCORES.DAT files.



CHALLENGE VS. DUEL

SINGLE-PLAYER CHALLENGE

- **The Mission:** Reach 5 full Barn Tokens to secure a victory.
- **Strikes Mechanic:** The player has a maximum of 3 lives, visually tracked on the HUD as colored heart-blocks.
- **Risk:** Reaching 3 wrong answers immediately triggers a GAME OVER state

TWO-PLAYER FARM DUEL

- **Local Multiplayer:** Allows two children to compete head-to-head on the same keyboard using alternating turns.
- **Sudden Death:** If both players hit the token ceiling simultaneously, the system enters an automatic sudden-death loop until a clear leader emerges.

BACKGROUND MUSIC ENGINE

- **Bare-Metal Audio:** Features a custom sound system that bypasses heavy drivers to program the built-in PC Speaker directly via internal I/O ports.
- **Timer Hook:** Utilizes an INT 8h hardware timer hook to cleanly advance through a retro melody table without interrupting active gameplay ticks.

